

END SEMESTER ASSESSMENTS, SPRING2024

Program: B.TECH COMPUTER SCIENCE & ENGINEERING/ B.TECH COMPUTER SCIENCE & BUSINESS SYSTEMS/ B.TECH COMPUTER SCIENCE & ENGINEERING (LATERAL)/ B.TECH COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)/ B.TECH COMPUTER SCIENCE AND ENGINEERING (IOT, CYBERSECURITY AND BLOCKCHAIN TECHNOLOGY)	
Subject Name: OPERATING SYSTEMS	
Subject Code: 1000013101/ 1000023101/ 1000033101/ 1000103101/ 1000113101	Semester: IV
Duration: 2 Hours	Maximum Marks: 50

Note: Answer the following questions as directed. Candidates can use non-programmable calculator.

Group A

(Answer All questions)

(Each question carries 1 mark)

1. Write the correct answer from the options given: (10 X 1)

- (i) Which of the following types of operating systems allow multiple users to access the system simultaneously?
- (a) Batch operating system (b) Multi-programmed operating system
- (c) Time-sharing operating system (d) Real-time operating system
- (ii) Scheduling a process from Ready Queue to CPU is done by
- (a) Short Term Scheduler (b) Middle Term Scheduler
- (c) Long Term Scheduler (d) Dispatcher.
- (iii) The general structure of a process consists of
- (a) Critical section (b) Reminder section
- (c) Race condition (d) Both (a) and (b).
- (iv) Page fault occurs when
- (a) The page is corrupted by application software (b) The page is not in main memory
- (c) The page is not in cache memory (d) One tries to divide a number by 0.

- (v) Variable partition memory allocation can lead to
- | | |
|----------------------------|----------------------------|
| (a) External fragmentation | (b) Internal fragmentation |
| (c) Both (a) and (b) | (d) None of these. |
- (vi) If a process has 32k bytes logical address space and the page size is 2048 bytes then the number of pages of that process is
- | | |
|--------|--------|
| (a) 4 | (b) 8 |
| (c) 16 | (d) 32 |
- (vii) Which system call is used to change the permissions of a file?
- | | |
|-------------|--------------|
| (a) chmod() | (b) write() |
| (c) ioctl() | (c) chperm() |
- (viii) Consider a system having 'n' resources of same type. These resources are shared by 3 processes: A, B, C. These have maximum demands of 3, 4, and 6 respectively. For what value of 'n' deadlock won't occur
- | | |
|--------|--------|
| (a) 15 | (b) 9 |
| (c) 10 | (d) 11 |
- (ix) What are system calls?
- | | |
|---|--|
| (a) Functions provided by the operating system to interact with hardware | (b) Functions provided by the compiler to execute system-level tasks |
| (c) Functions provided by the application software to communicate with other applications | (d) Functions provided by the network protocols for data transmission. |
- (x) In a Resource Allocation Graph (RAG), what do the nodes represent?
- | | |
|---------------|---------------|
| (a) Processes | (b) Resources |
| (c) Deadlocks | (d) Threads |

Group B

(Answer any Four questions out of Six)

(Each question carries 5marks)

(4 X 5)

2. Draw the process state diagram and explain different states.

3. Consider the following set of processes, with the length of the CPU-burst time given in millisecond:

Process	Arrival Time	Burst Time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

- (a) Draw two Gantt charts to illustrate the execution of these processes for the preemptive shortest job first (SJF) and round robin (RR) (time quantum = 4) scheduling.
- (b) Calculate the average waiting time and average response time for both the scheduling strategies. $[3 + (1+1)]$
4. (a) Define the terms 'race condition' and 'critical-section'.
- (b) A counting semaphore S is initialized to 17. Then, 21 P operations and 13 V operations are performed on S. What is the final value of S? $(2 + 3)$
5. (a) If the hit ratio to a TLB is 80%, and it takes 15 nanoseconds to search the TLB, and 150 nanoseconds to access the main memory, then what must be the effective memory access time in nanoseconds?
- (b) Differentiate between internal and external fragmentation. $(2 + 3)$
6. (a) Write down the merits and demerits of a virtual memory system.
- (b) A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references

(reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1. If optimal page replacement policy is used, how many page faults occur for the above reference string?
(2 + 3)

7. Discuss about multi-programmed and time sharing operating system with example.
(1 + 4)

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. Consider the following system snapshot using data structures in the banker's algorithm with resources A, B, C and D and processes P0 to P4. Resource type A has 3 instances, B has 17 instances, C has 16 instances, and D has 12 instances.

Given Matrices												
Process	Allocation Matrix				Max Matrix				Available Matrix			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	1	1	0	0	2	1	0	1	5	2	0
P1	1	2	3	1	1	6	5	2				
P2	1	3	6	5	2	3	6	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

- (a) Is this system currently in a safe state? If it is safe then write the safe sequence.
(b) If a request from P2 arrives for (1, 0, 0, 1), can that request be safely granted immediately?
(5 + 5)
9. (a) Describe the first fit, best fit and worst fit strategies with suitable example. Write their merits and demerits for memory space allocation.

(b) Define thrashing with suitable diagram. How does it affect performance?

[6 + (2 + 2)]

10. Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, and 70. Assume that the initial position of the read / write head is on track 50. Find the additional tracks that will be traversed by the read / write head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves outwards when it starts execution).

11. Write short notes of the following: (Any Two)

(5 + 5)

- (a) Context switching
- (b) Swapping
- (c) Inode
- (d) Segmentation.